Phase 12 SOTA CVRP HGS Baseline Results (2026-07-21)

This note records the official external HGS-style CVRP calibration on branch sota-cvrp-hgs-baseline.

The official artifact archive is under runs/cvrp_phase12_sota_hgs_baseline/pyvrp_hgs_20260721_heldout_30sx5.

Protocol

- Solver: PyVRP 0.13.4, using its HGS-style vehicle-routing solver.
- Dependency parser: VRPLIB 2.2.0.
- Panel: the same five held-out CVRPLIB X instances used in phase 9 and the phase-9 closeout.
- Seeds: 1, 2, 3, 4, 5.
- Runtime cap: 30 seconds per seed.
- Reporting rule: best validated seed per instance.
- Validation: every selected route set was revalidated with the project phase-9 CVRP validator.

Results

Instance	Best seed	Cost	Best known	Gap	Wall seconds
X-n176-k26	5	48161	47812	0.007299	30.036
X-n190-k8	4	17020	16980	0.002356	30.042
X-n223-k34	4	40726	40437	0.007147	30.056
X-n247-k50	5	37671	37274	0.010651	30.057
X-n275-k28	3	21320	21245	0.003530	30.058

The mean held-out penalized gap for all 25 runs is 0.008779. The mean held-out penalized gap for the best validated seed per instance is 0.006197, with feasibility rate 1.0.

Validation Audit

The archive was audited after the run:

- all 25 expected runs are present,
- all five held-out instances have one selected best seed,
- all selected best routes revalidate as feasible with the phase-9 validator,
- the recomputed best-seed mean held-out penalized gap is 0.006197,
- the selected best routes match the costs and gaps stored in the CSV and JSON summaries.

Interpretation

This calibration strengthens the thesis boundary. PyVRP's best-seed mean held-out gap of 0.006197 is lower than the Clarke-Wright mean held-out gap of 0.073766 and far lower than the phase-9 closeout replay-evolution mean held-out gap of 0.171114.

The calibration does not invalidate the phase-9 mechanism result because PyVRP is an external specialized solver, not a learned code-evolution condition. It does rule out a practical claim that the evolved CVRP solvers are competitive with mature HGS-style vehicle-routing search on this held-out benchmark slice.

The thesis should therefore use this phase as an external baseline calibration: current LLM-guided code evolution can produce feasible CVRP solver logic and mechanism-level improvements over weaker learned controls, but mature specialized VRP search remains the stronger reference point.

Sources

- Protocol: docs/PHASE_12_SOTA_CVRP_HGS_BASELINE_PROTOCOL.md
- Runner: run_cvrp_phase12_sota_hgs_baseline.py
- Result summary: runs/cvrp_phase12_sota_hgs_baseline/pyvrp_hgs_20260721_heldout_30sx5/summary.md
- Aggregate JSON: runs/cvrp_phase12_sota_hgs_baseline/pyvrp_hgs_20260721_heldout_30sx5/summary.json
- Per-seed rows: runs/cvrp_phase12_sota_hgs_baseline/pyvrp_hgs_20260721_heldout_30sx5/pyvrp_hgs_runs.csv
- Best-by-instance rows: runs/cvrp_phase12_sota_hgs_baseline/pyvrp_hgs_20260721_heldout_30sx5/pyvrp_hgs_best_by_instance.csv
- Best routes: runs/cvrp_phase12_sota_hgs_baseline/pyvrp_hgs_20260721_heldout_30sx5/best_routes.json